DigitalTwin.ai

Predicting where bottlenecks form and which vehicles carry defects — before the vehicle reaches the station.

TEAM

Combinators

IIT (ISM) Dhanbad

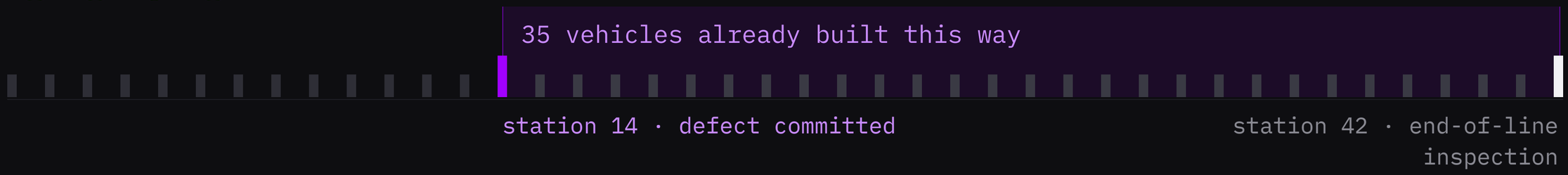
Yash Saxena Sahil Thakur Nipun Shekhar

By the time the line finds the defect, 35 vehicles carry it.

A defect committed early in the line does not surface until end-of-line inspection. By then dozens of units have passed the same station under the same conditions and carry the same issue.

DETECTION PROBABILITY	
0.15	in-station self-check
0.45	mid-line
0.92	end-of-line

35.0 vehicles — mean exposed cohort per escaped defect measured on the simulated line



42-station reference line

Four constraints, not four features.

C1

Sensor coverage is a patchwork, not a grid.



23 full 10 partial 9 dark — time-stamps only

C2

Root causes are multi-causal and intermittent.

Equipment wear, operator variation, incoming lot quality, ambient conditions — rarely one of them alone, rarely twice the same way.

C3

Live production systems cannot be modified.

Changes land in rare maintenance windows only. Anything that requires touching the line to work does not get deployed.

C4

False alarms destroy floor trust faster than true positives build it.

Adoption is the binding constraint on value, so alert volume is a design parameter — not an output.

Time pressure couples bottlenecks to defects — and it is calculable in advance.

$$\text{pressure}(\text{station}, \text{vehicle}) = \frac{\text{required_work_content}(\text{station}, \text{variant})}{\text{available_time}(\text{station}, \text{vehicle})}$$

numerator	The process plan. Known before the vehicle physically exists.
denominator	From forward simulation: takt, minus blocking from downstream buffer saturation, minus schedule recovery after an upstream stop, plus idle from starving.
above 1.0	The operator must rush, or the line must stop.

Build sequence is published 90 minutes ahead

DEFECT-BIAS VS PRESSURE



Flat below 0.90, steep above it. A lagging cycle-time control chart only moves after the cycle has already run long — by which point the vehicle is built. A forward simulation sees the excursion coming.

Two simulators, strictly separated.

Ground truth

src/twin/plant/

operator fatigue
tool wear
lot deviation
ambient drift
operator skill
true margin-to-spec
probabilistic inspection

Emits a masked, noisy, partial telemetry stream.
Never shipped as product.

masked / noisy
partial telemetry only

one arrow · one direction

The product

src/twin/kernel/

state estimate — from telemetry only
its own independent forward simulation
the pressure definition

No access to anything on the left.

tests/test_isolation.py

Walks the abstract syntax tree of every module that ships as product and fails the build if any of them imports the ground-truth simulator, references it as a string, or opens one of its artefacts. Without it, every result in this deck would be self-graded.

The pipeline.

01 Telemetry ingest
OPC-UA-shaped

→

02 State estimate

→

03 Forward simulation
90-min horizon

→

04 Channel features

→

05 Quantile risk model

→

06 PFMEA expected-cost ranking

→

07 Three persona views

DESIGN NOTE 01

Regress normalised margin-to-specification, not a binary defect label.

Defects are 0.065% of station visits; a classifier spends its capacity learning the base rate. Margin is dense — every visit carries a real value — and a seven-quantile spread gives an uncertainty estimate for free.

P(defect) = P(margin < 0), then isotonically calibrated on a held-out slice.

DESIGN NOTE 02

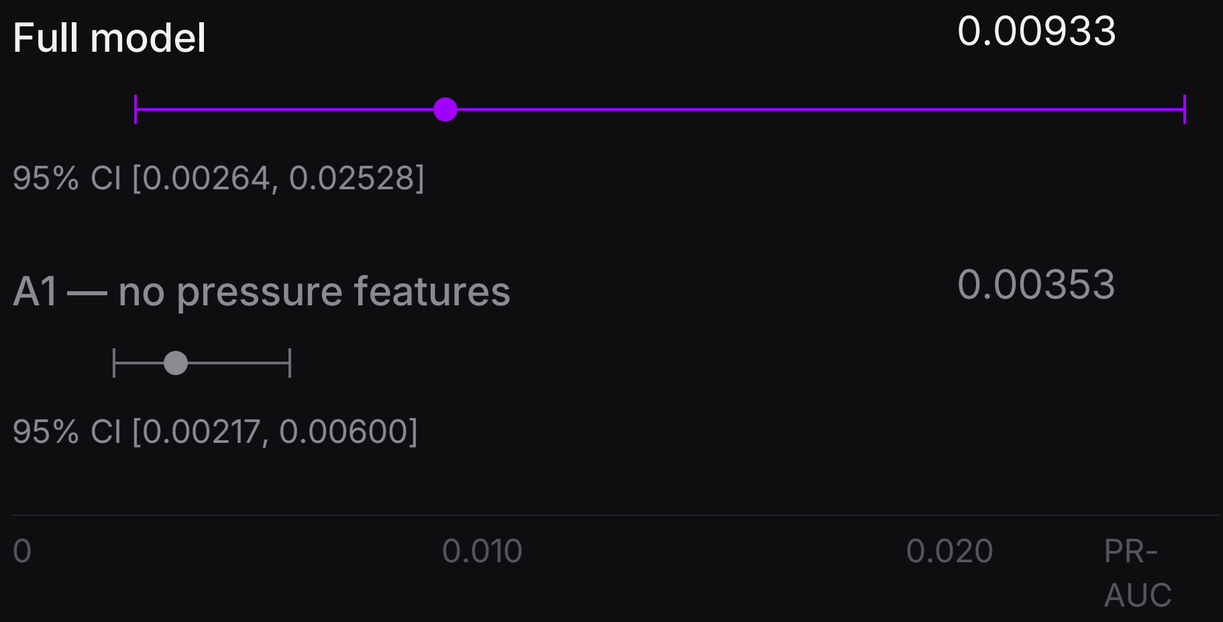
Graceful degradation is mechanical, not a fallback path.

The gradient-boosting model handles NaN natively, so a dark station's missing channels go straight in. Channel-block dropout during training is what stops the model leaning on instrumentation it will not always have.

The ablation that could have refuted us.

−62.2%

PR-AUC when every pressure-derived feature is removed and everything else is refitted



Fine print: the numerator of pressure — planned work content — is deliberately left in. So this is a clean test of whether the forward-simulated available-time denominator adds anything beyond knowing how much work the variant needs.

“The Round 1 thesis survived an experiment designed to be able to refute it. Had it not, our results file would say so.”

Held-out test days · bootstrap intervals

Against the baselines.

test days · base rate 0.0650%
159,962 station visits · 104 defects

MODEL	PR-AUC	LIFT	PREC@10/SHIFT	LIFT
DigitalTwin.ai	0.00933	14.3x	0.037	58x
B1 — Western Electric SPC	0.00086	1.3x	0.000	—
B0 — base rate	0.00081	1.2x	0.000	—

58x, plainly: of the ten station visits flagged this shift, one is 58 times more likely to carry a real defect than ten picked at random.

LEAD TIME BEFORE COMMIT

23.1 min median

p10: 19.0 min — about 23 vehicles of warning

CALIBRATION

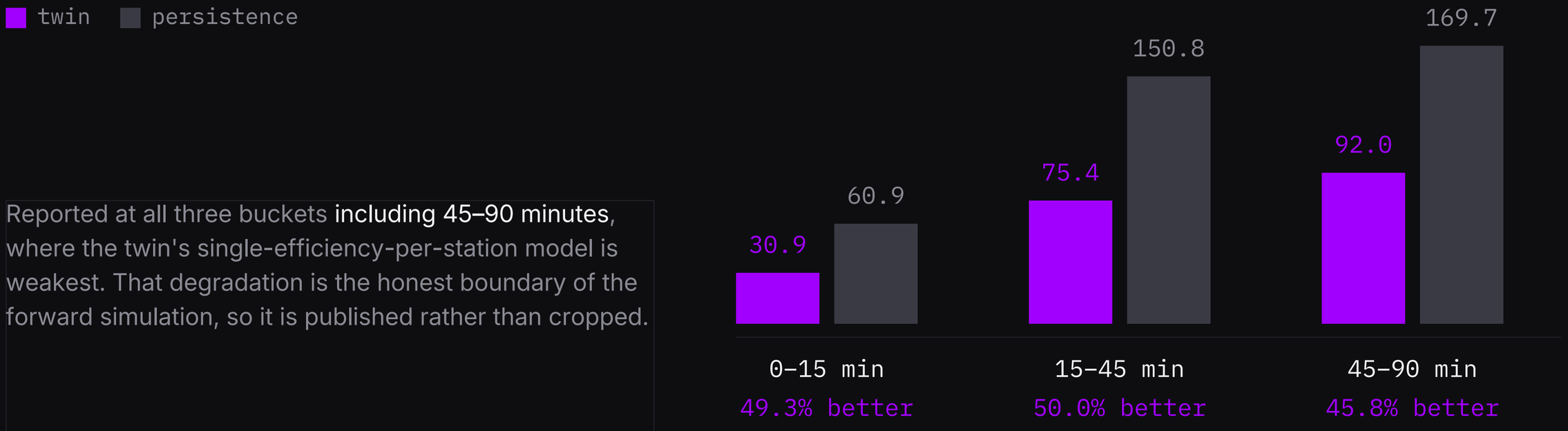
Expected Calibration Error 0.0004 on test days the isotonic calibrator never saw. Brier 0.000650.

HONESTY

PR-AUC is noisy at this base rate. Around a hundred positives; per-day values span two orders of magnitude. Intervals are percentile bootstraps over 300 row resamples. The point estimates are real. They are not precise.

Is it actually a twin?

The twin's own forward simulation predicting station arrival times, scored against a naive persistence baseline. Mean absolute error, in seconds.



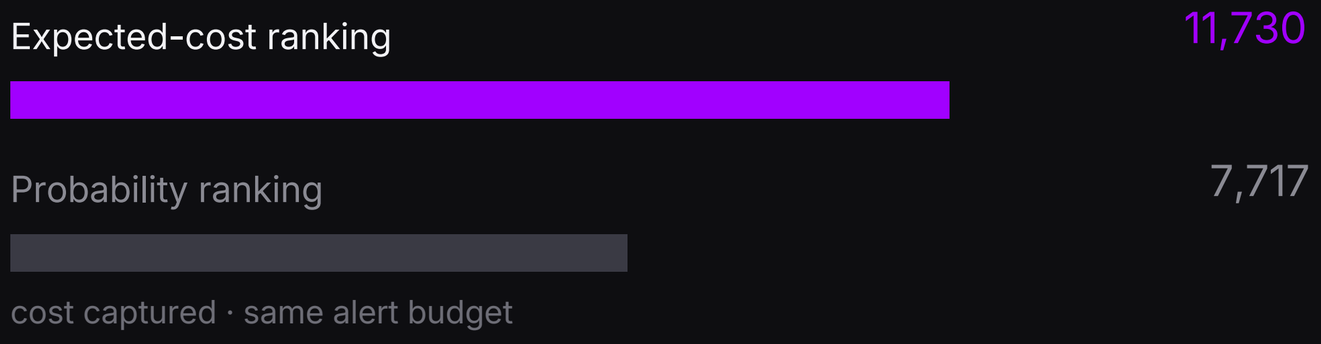
Reported at all three buckets including 45–90 minutes, where the twin's single-efficiency-per-station model is weakest. That degradation is the honest boundary of the forward simulation, so it is published rather than cropped.

Arrival-time MAE, seconds

From probability to decision.

expected_cost = P(defect) × Σd P(escape to detection point d) × costd × severity_weight

Ablation A4 — identical alert budget, identical model, identical probabilities.
The only change is the ordering.



+52.0%

uplift from ordering alone

Grade A · act now

Grade B · evaluate

Grade C · monitor

Thresholds fitted on the calibration slice and frozen, so grades on test days are not defined by the test days' own distribution.

The system recommends. The human decides. There is no write path to the line.

One click, backwards along the line.

The direct answer to “a defect introduced early may not surface until much later, by which time many downstream units carry the same issue.”



One click on an end-of-line defect returns **every VIN that passed the implicated station inside the drift window**, each with a risk score and its current position on the line.

Verified against plant ground truth in a test — the returned cohort is checked against the units the ground-truth simulator actually exposed.

Schematic · units drawn to illustrate the window, not measured counts

Three personas, one model.

/floor — real time

Floor supervisor

Live line map coloured by current pressure.

Forward pressure heatmap for the next 90 minutes.

Ranked alert queue with one-click cohort trace.

DECISION

Which station to walk to next.

/manager — shift & week

Plant manager

Pressure aggregated by station × hour.

Recurring loss contributors by defect class.

Sequencing what-if with before/after pressure comparison.

DECISION

What to change in the schedule.

/leadership — investment case

Leadership

ROI derived from measured containment and PFMEA costs.

Rollout tiers by sensor maturity.

Portability to a second plant.

DECISION

Whether and where to fund rollout.

One model, three aggregations. Every displayed number carries a provenance marker showing which computation produced it.

What is measured, and what is assumed.

MEASURED — HELD-OUT TEST DAYS

14.3x PR-AUC lift over base rate; **58x** precision lift at ten alerts per shift.

23.1 min median warning before commit.

+52.0% cost captured from expected-cost ranking at identical alert budget.

Cohort containment: **1.0%** at 10 alerts/shift, **3.2%** at 100 alerts/shift.

ASSUMED — NOT MEASURED

250 working days per year.

PFMEA costs are directional, in unnamed currency units.

Containment converts to money 1:1 — **the weakest assumption in the business case.** The containment rate is measured on held-out days; the exposure it multiplies is a model of a simulated plant.

ALERTS/SHIFT	PRECISION	RECALL	CONTAINMENT
10	0.037	0.029	1.0%
25	0.020	0.038	1.2%
50	0.010	0.038	2.4%
100	0.006	0.048	3.2%

alert-budget sweep · published in full

Containment at ten alerts a shift is limited by the **alert budget**, not by the model — ten alerts against a 0.065% base rate cannot reach most defects. **This is our weakest measured result and the business case leans on it. It is the number to challenge first.**

Each phase has an entry condition and an exit gate.

phases 1 and 2 are sized by measured ablations, not estimates

PHASE 0

Shadow mode

Point the twin at a historian, score it against outcomes that already happened. No alerts reach the floor.

exit gate

A full production cycle of validated forecasts.

False alarms erode floor trust faster than they build it.

PHASE 1

One line, dark start

Nothing but entry and exit timestamps. Establishes the floor of usefulness before any instrumentation spend.

sized by

A2 — measured −39.7%

Against the full model. A measurement, not a projection.

PHASE 2

Targeted retrofit

Cameras at sensor-poor stations. Retrofits land in scheduled maintenance windows only.

sized by

A3 — measured −48.4%

What the retrofit buys, measured before it is bought.

PHASE 3

Second line, second plant

34 stations, 72-second takt, 40% dark.

the entire change

Loading that file is the whole portability change — no code moves.

Filled node = phase sized by a measured ablation

Risk, and the mechanism that answers it.

RISK	MITIGATION
Alert fatigue destroys adoption	Fixed per-shift alert budget; ranking by expected cost, not probability; the full budget sweep is published so the trade-off is explicit rather than tuned in secret.
Model looks good because it cheated	tests/test_isolation.py enforces simulator separation at the AST level and fails the build on breach.
Metrics are noisy and could mislead	Bootstrap 95% intervals on every estimate; run-to-run variance disclosed in LIMITATIONS 12b rather than hidden.
Instrumentation may never arrive	Model handles NaN natively; channel-block dropout during training; A2 measures the fully-dark floor.
Automated action damages the line	No write path exists. Recommend-only by design, with Grade A/B/C decision bands for a human.
Plant-to-plant variation	Layout, takt, coverage and buffers are all config; portability demonstrated on a second line profile.
Predictions unvalidated against reality	Shadow-mode protocol as Phase 0, gated on a full production cycle.

What this prototype does not do.

☐ No vision inference	The vision channel is a documented stub with a real interface — item 1 in our limitations file, and said out loud in the demo.
☐ No real OPC-UA connection	An abstract source with a replay implementation.
☐ No real plant data	A simulated plant with tuned constants, every one of them listed in the assumptions file.
☐ No causal attribution	Beyond four defect classes, nothing is claimed.
☐ No write path to the line	By design, not by omission.
☐ No operator-fatigue state in the forward model	And we measured what that costs us.

Everything above is in docs/LIMITATIONS.md, written before this deck existed.

Time pressure per station per vehicle is the coupling variable between bottlenecks and defects and because the build sequence is published 90 minutes ahead, that pressure is calculable before the vehicle arrives.

REPRODUCE IT YOURSELF

```
$ pip install -r requirements.txt
$ python make.py reproduce
$ python make.py demo
```

fixed seed · CPU only · no GPU
no network access at any point · no API key

Every number in this deck is regenerated by that command.